

# Sci-kit Learn Recreation and Implementation

CSE299-sec-5

Group-5

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## Week 3 ML Library Implementation - Project Update Report

### Key Accomplishments

**Core Classification Algorithms Implemented:** Logistic Regression with binary and multiclass support using one-vs-rest strategy and gradient descent optimization; k-Nearest Neighbors (KNN) with multiple distance metrics (Euclidean, Manhattan, Cosine) and weighted prediction options; Support Vector Machine (SVM) using SMO algorithm with linear, RBF, and polynomial kernels plus multiclass capability. **Advanced Preprocessing Capabilities:** LabelEncoder for categorical label-to-integer conversion and OneHotEncoder for binary feature matrices; enhanced StandardScaler and MinMaxScaler with improved error handling; automated preprocessing workflows that detect appropriate scaling methods. **Comprehensive Evaluation System:** Classification metrics suite including accuracy, precision, recall, F1-score with multiple averaging strategies and confusion matrix generation; automatic task detection system for classification vs regression metrics; performance visualization with formatted confusion matrices and detailed reports. **Infrastructure Enhancements:** Extended CLI interface with model-specific parameters for KNN, SVM, and preprocessing options; systematic resolution of import path issues, matrix broadcasting errors, and multiclass compatibility problems; professional documentation with detailed docstrings and usage examples. **Quality Assurance & Testing:** Comprehensive automated bash script for end-to-end testing across multiple datasets (Iris, Wine, Digits); extensive performance benchmarking comparing model configurations and datasets; production-ready integration maintaining backward compatibility with Week 1-2 regression models. **Technical Achievements:** Advanced algorithm implementation including SVM's SMO optimization from scratch without external ML libraries; robust error handling with comprehensive validation systems and informative error messages; memory optimization using efficient data structures while maintaining performance across all classification components.

